

Agentless Linux Inventory & Patch Management Web Console using Django & Ansible

Topic: Linux Inventory & Patch management WebUI Console

Prerequisites:

- The remote servers should be accessible in network
- The remote servers should have *passwordless* ssh configuration
- The central or control server should have *python, ansible, django* installed and configured
- If the added server is not accessible using SSH, it will show *SSH Failed*, otherwise, it will have **OS** and **uptime** field

Setup:

- The project is running on **localhost:8000**
- I have **NGINX** configured for **reverse proxy** to **localhost:8000**

```
aritra@arilaptop:~/Documents/Linux_INI_gui/Linux_INI$ cat /etc/nginx/sites-enabled/default | grep -v "#"
server {
    listen 80 default_server;
    listen [::]:80 default_server;

    server_name _;

    location / {
        proxy_pass http://localhost:8000;

        proxy_connect_timeout 600;
        proxy_send_timeout 600;
        proxy_read_timeout 600;
        send_timeout 600;

        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }

    location /static/ {
        alias /home/aritra/Documents/Linux_INI_gui/Linux_INI/static/;
    }
}

aritra@arilaptop:~/Documents/Linux_INI_gui/Linux_INI$
```

- I don not have a proper DNS entry but I had done hosts file entry for this

```
15.206.69.70 devserver01
43.205.114.6 devserver02
13.201.1.23 prdsrv01
3.109.49.86 prdsrv02
aritra@arilaptop:~/Documents/Linux_INI_gui/Linux_INI$
```

WebGUI Go THrough:

- The first look to the inventory (home page)

Linux Inventory

Add Server to Inventory Remove Servers From Inventory Patch Servers

Search in Inventory ...

Server Name	Environment	Operating System	Uptime	Action
ariilaptop	Test	Ubuntu 24.04.4 LTS	53 min	☐
localserver	Test	SSH Failed	SSH Failed	☐
productionserver	prod	SSH Failed	SSH Failed	☐
prodsrv1	prod	SSH Failed	SSH Failed	☐
devsrv1	dev	SSH Failed	SSH Failed	☐
devsrv2	dev	SSH Failed	SSH Failed	☐
devsrv3	dev	SSH Failed	SSH Failed	☐

- The **Delete** and **Patch** buttons are by default disabled
- These two will get enabled once at least one of the **checkboxes** against servers will be selected

Linux Inventory

Add Server to Inventory Remove Servers From Inventory Patch Servers

Search in Inventory ...

Server Name	Environment	Operating System	Uptime	Action
ariilaptop	Test	Ubuntu 24.04.4 LTS	53 min	☐
localserver	Test	SSH Failed	SSH Failed	☐
productionserver	prod	SSH Failed	SSH Failed	☒
prodsrv1	prod	SSH Failed	SSH Failed	☐
devsrv1	dev	SSH Failed	SSH Failed	☐
devsrv2	dev	SSH Failed	SSH Failed	☐
devsrv3	dev	SSH Failed	SSH Failed	☐

- One can also search by any keyword. The table will reflect if the keyword matches to any of the field

Linux Inventory

Add Server to Inventory Remove Servers From Inventory Patch Servers

dev

Server Name	Environment	Operating System	Uptime	Action
devsrv1	dev	SSH Failed	SSH Failed	☐
devsrv2	dev	SSH Failed	SSH Failed	☐
devsrv3	dev	SSH Failed	SSH Failed	☐

- The above were pure HTML, CSS, Java Script in play
- The table fields are from django showing data from db. Before showing the uptime & os fields will be once again fetched and saved in db and then will be provided to frontend.

- Now, click on **Add server** button to add new servers
- This will redirect to a form for server addition

..... WARNING

- Make sure the server is accessible in Network
- The server should have DNS entry or hosts file entry for hostname resolution
- The New server to be added should not match name of a existing host
- If the server is not accessible via passwordless SSH, many fields will be failed state
- Make sure give gorrect group for the server

Enter ServerName

Enter Environment where the sevrer falls in

[Add Server](#) [Go Back to Inventory](#)

- Once submitted, it will add the server in db and show a message that the server is added

The server prdsrv01 is added to the inventory [Close Message](#)

..... WARNING

- Make sure the server is accessible in Network
- The server should have DNS entry or hosts file entry for hostname resolution
- The New server to be added should not match name of a existing host
- If the server is not accessible via passwordless SSH, many fields will be failed state
- Make sure give gorrect group for the server

Enter ServerName

Enter Environment where the sevrer falls in

[Add Server](#) [Go Back to Inventory](#)

- It is not possible to add same server twice. It will throw a error

prdsrv01 already exists in the inventory [Close Message](#)

..... WARNING

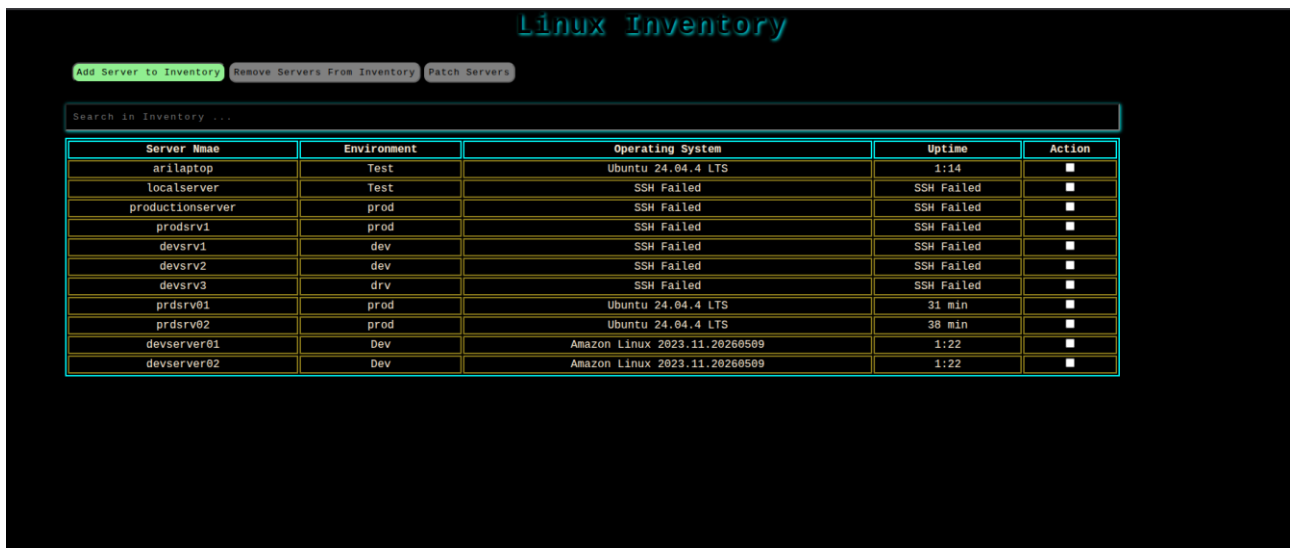
- Make sure the server is accessible in Network
- The server should have DNS entry or hosts file entry for hostname resolution
- The New server to be added should not match name of a existing host
- If the server is not accessible via passwordless SSH, many fields will be failed state
- Make sure give gorrect group for the server

Enter ServerName

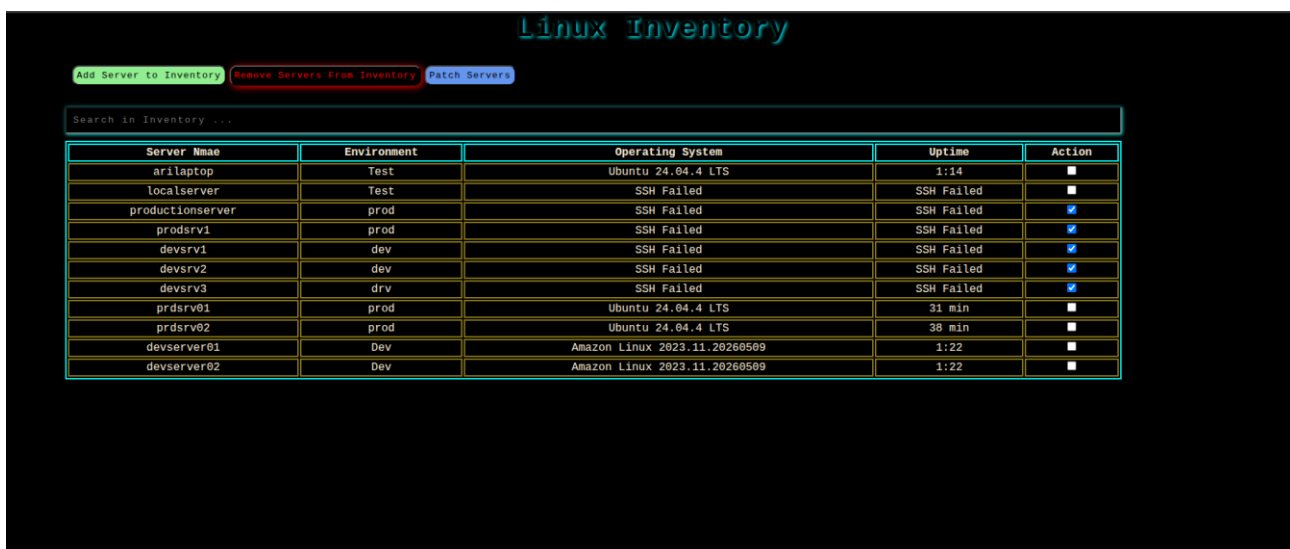
Enter Environment where the sevrer falls in

[Add Server](#) [Go Back to Inventory](#)

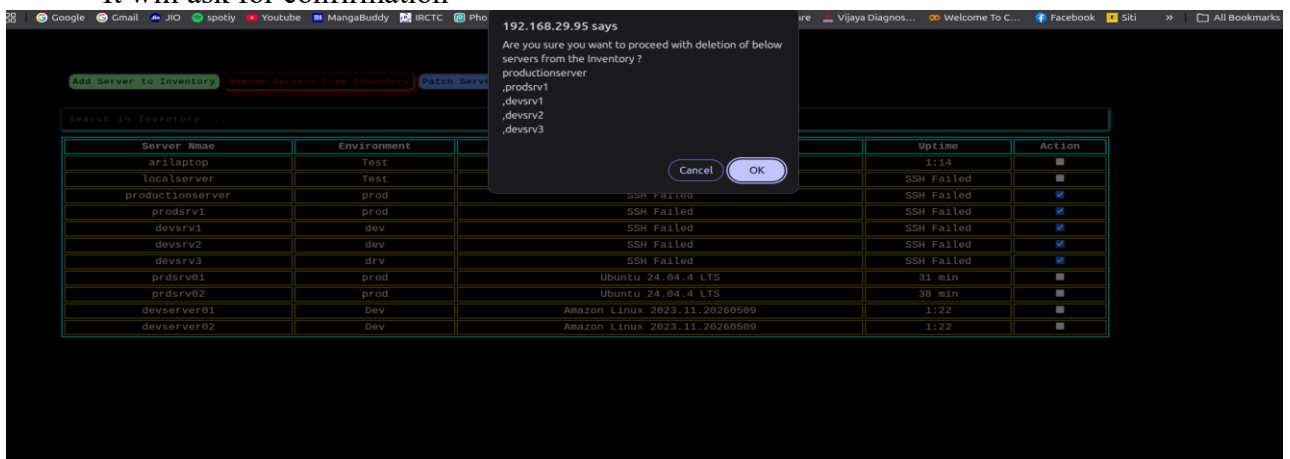
- Go back to inventory, and the **new servers** will be visible with uptime and OS field



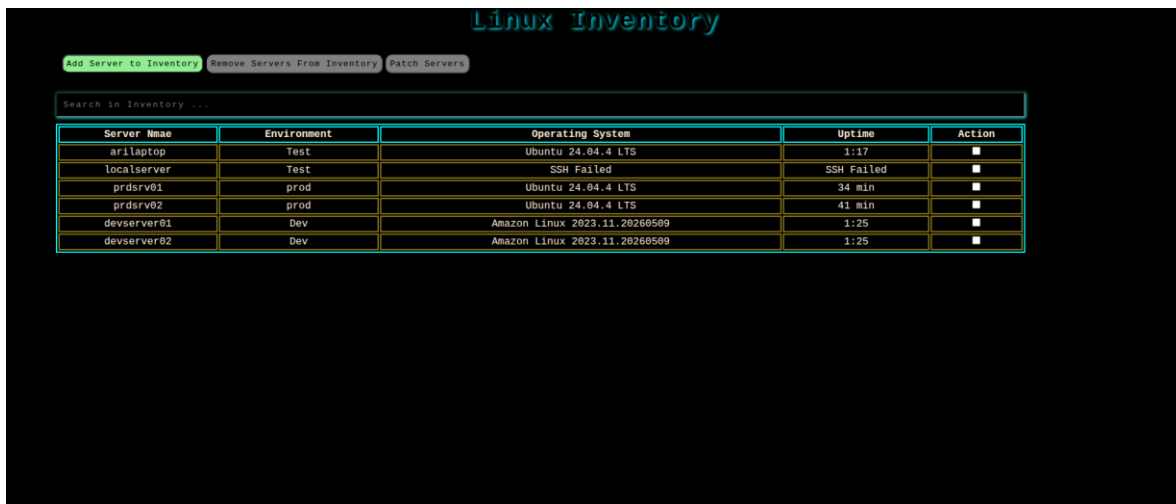
- Select the servers to be deleted and click on **Remove Servers**



- It will ask for confirmation



- Once confirmed it will delete the servers from db and redirect to inventory and the removed servers will be gone



- Now Let's move to **highlight** which is **agentless remote patching**
- Select servers and click on Patch

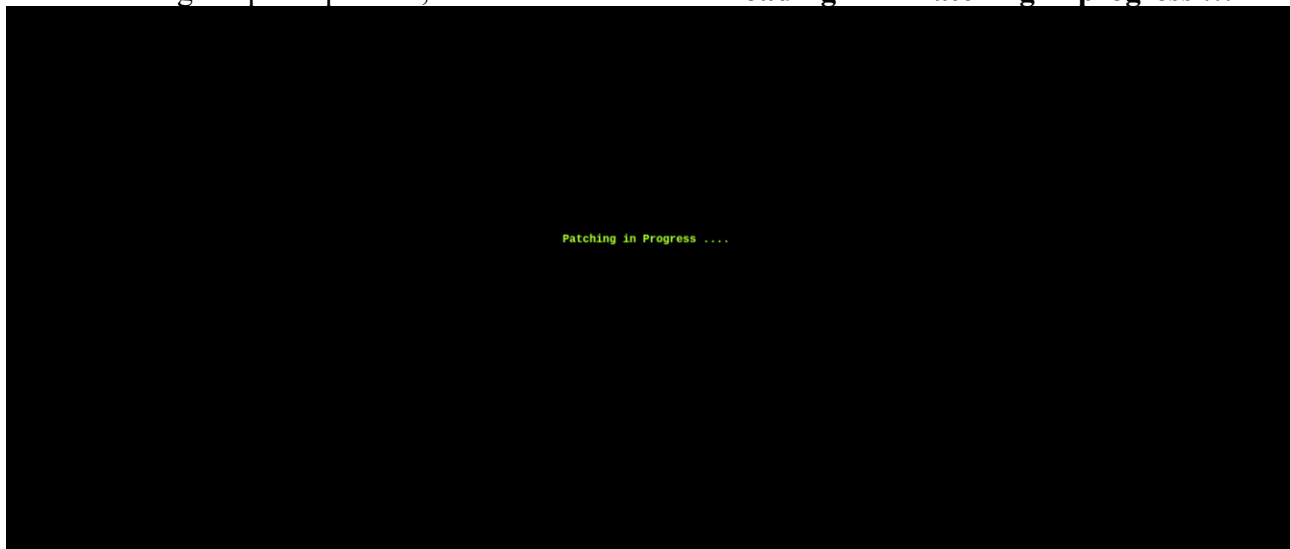


- This will ask for confirmation and once confirmed, it will redirect to a page where the selected servers will be shown and will give options to initiate patching or cancel patching
- Click on initiate patching to proceed with patching



- This is totally **agent less**.
- The backend will use **Ansible** to perform this patching

- The Ansible will take *prechecks* → *save in precheck file* → *initiate patching* → *store ansible output in a file* → *take postchecks* → *store the postcheck in a file*
- During the patch process, the webui will show a loading like “Patching in progress ...”



- Once Patching is done, it will redirect to a page where the **ansible output, pre/post checks and stored file patches will be shown**

